Palak Deb Patra

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palakdebpatra CaptainBlue793 BlueMask6

EDUCATION

National Institute of Technology, Rourkela — **Dec 2021 – Jun 2025**
Bachelor of Technology in **Computer Science & Engineering** | **CGPA**-8.52/10 — **Rourkela, India**

RESEARCH EXPERIENCE

NIT Rourkela | **Undergraduate Student Researcher** — **Dec 2024 – Mar 2025**
Python, PyTorch, GAN, Conv-GRU — **Hybrid, India**

- Created and preprocessed a custom dataset with unpaired images from **LoL** and **SICE** datasets.
- Applied normalization and data augmentation techniques for robust model training.
- Developed a novel neural network architecture, the **Bidirectional Conv-GRU integrated CycleGAN**, for low-light image enhancement on unpaired datasets.
- Attained results with **PSNR = 20.90 dB**, **SSIM = 0.74**, and **LPIPS = 0.22**, showing improved brightness, contrast, and detail preservation over baseline models.

PROFESSIONAL EXPERIENCE

NatWest Group | **Software Engineering Intern** — **May 2024 – Jul 2024**
Python, GitLab, GraphQL, RESTful API, Snowflake, ThoughtSpot — **Gurgaon, India**

- Utilized **GitLab's GraphQL API** and **REST API** to collect organization-wide DevOps metrics and KPIs for performance tracking.
- Automated data retrieval to process paginated JSON responses and export results to CSV, reducing processing time by **28%**.
- Uploaded **16 CSV datasets** to **Snowflake**, designed a unified data model, and integrated with **ThoughtSpot** to create interactive dashboards.

PROJECTS

MeLoVerse | Python, Colab, spaCy, Streamlit, Scikit-learn | Link | Code — **Mar 2024**

- Developed a **music recommendation system** using **10,000** Spotify tracks to generate closely matched recommendations based on user preferences.
- Integrated the **Spotify Developer API** to retrieve metadata and album art, enhancing the UI with dynamic and contextually relevant visuals.

SummaScribe | Python, Streamlit, LangChain | Link | Code — **Aug 2023**

- Built a web application using **LLMs (LaMini)** and NLP techniques to **summarize text documents**, enabling rapid extraction of key insights from PDFs.
- Implemented dynamic summary generation producing concise outputs between **50–1000 words** based on user preference.

SKILLS

Coursework	DSA, DBMS, OOP, Machine Learning, NLP, Computer Vision, Deep Learning, Computer Networks
Languages	Python, C++, Java, TypeScript, MySQL, GraphQL
Frameworks	PyTorch, TensorFlow, Scikit-learn, Streamlit, LangChain, spaCy, NLTK, Spring Boot, Angular
Developer Tools	Git, GitHub, GitLab, HuggingFace, Postman
Cloud & Deployment	AWS (S3, EC2, RDS), Snowflake, Oracle

ACHIEVEMENTS

- Selected for the **Amazon ML Summer School 2023**, collaborating with data scientists on advanced ML topics.
- Solved **800+** problems on **LeetCode** with a peak rating of **1795**, ranking in the **top 8%** globally.
- Secured global ranks of **277** in Weekly Contest 438 and **1839** in Weekly Contest 437 on **LeetCode**.
- Earned a **perfect (100%) score** in the **LLMs and Prompt Patterns Utilization** program by Vanderbilt University.

EXTRACURRICULAR ACTIVITIES

Coordinator | Event Management | **NITRUTSAV 2023** — **Aug 2023 – Sep 2023**

- Led a team of 7 volunteers to execute NIT Rourkela's flagship cultural festival, securing a **20%** sponsorship increase and boosting participant engagement by **11.26%**.

Senior Staff Member | Social Service | **AASRA** — **Jun 2022 – Jan 2024**

- Contributed to the DIYA Project, helping raise **Rs 2,11,000** in funds and generate **Rs 1,32,000** in profit through coordinated outreach and fundraising efforts.